Evaluating Text Classification

We need a metric for knowing how well our spam detector (or pie-tweet-detector) is doing. To evaluate any system for detecting things, we start by building a confusion matrix

		gold standard labels		
		gold positive	gold negative	
system output labels	system positive	**true positive**	**false positive**	**precision** $= \frac{tp}{tp+fp}$
	system negative	**false negative**	**true negative**	
		recall $= \frac{tp}{tp+fn}$		**accuracy** $= \frac{tp+tn}{tp+fp+tn+fn}$

Figure 4.4 A confusion matrix for visualizing how well a binary classification system performs against gold standard labels.

Evaluating Text Classification

We have three metrics now: **accuracy**, **precision**, and **recall**… which one do we use?

NOT accuracy!! Why?

Imagine we had 1 million tweets, of which only 100 talk about our pies. That leaves 999,900 tweets that have nothing to do with our pies.

Now imagine we have a stupid classifier that always classifies everything as "not about pie"

This would give us **999,900 true negatives** and only **100 false negatives** for an accuracy of **999,900/1,000,000** or **99.99%**

But we know that this classifier is stupid and won't actually do well…

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Evaluating Text Classification

We have three metrics now: accuracy, precision, and recall… which one do we use?

So we generally turn to precision and recall instead

Precision measures the % of the items that the system detected (i.e., the system labeled as positive) that are in fact positive (i.e., are positive according to the human gold labels)

$$\textbf{Precision} = \frac{\text{true positives}}{\text{true positives} + \text{false positives}}$$

Evaluating Text Classification

We have three metrics now: accuracy, precision, and recall... which one do we use?

So we generally turn to precision and recall instead

Recall measures the % of items actually present in the input that were correctly identified by the system

$$\textbf{Recall} = \frac{\text{true positives}}{\text{true positives} + \text{false negatives}}$$

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Evaluating Text Classification

We have three metrics now: **accuracy**, **precision**, and **recall**... which one do we use?

Thus **precision** and **recall**, unlike **accuracy**, emphasize true positives: finding the things that we are supposed to be looking for

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F-Measure

How do we combine precision and recall into one metric?

One way is by defining the **F-measure**

$$F_\beta = \frac{(\beta^2 + 1)PR}{\beta^2 P + R}$$

The β parameter differentially weights the importance of recall and precision, based perhaps on the needs of an application.

Values of β > 1 favor recall, while values of β < 1 favor precision.

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F-Measure

How do we combine precision and recall into one metric?

One way is by defining the **F-measure**

$$F_\beta = \frac{(\beta^2+1)PR}{\beta^2 P + R}$$

When β = 1, precision and recall are equally balanced; this is the most frequently used metric, and is called $F_{\beta=1}$ or just F_1

$$\mathrm{F}_1 = \frac{2PR}{P+R}$$

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Evaluating with more than two classes

Okay so we covered binary classification… but what if we have >2 classes?

		gold labels			
		urgent	normal	spam	
system output	urgent	8	10	1	$\textbf{precision}_u = \frac{8}{8+10+1}$
	normal	5	60	50	$\textbf{precision}_n = \frac{60}{5+60+50}$
	spam	3	30	200	$\textbf{precision}_s = \frac{200}{3+30+200}$
		$\textbf{recall}_u = \frac{8}{8+5+3}$	$\textbf{recall}_n = \frac{60}{10+60+30}$	$\textbf{recall}_s = \frac{200}{1+50+200}$	

Figure 4.5 Confusion matrix for a three-class categorization task, showing for each pair of classes (c_1, c_2), how many documents from c_1 were (in)correctly assigned to c_2.

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Evaluating with more than two classes

Now we have 3 separate precisions and 3 separate recalls… how do we combine them into 1?

		gold labels			
		urgent	normal	spam	
system output	urgent	8	10	1	$\textbf{precision}_u = \frac{8}{8+10+1}$
	normal	5	60	50	$\textbf{precision}_n = \frac{60}{5+60+50}$
	spam	3	30	200	$\textbf{precision}_s = \frac{200}{3+30+200}$
		$\textbf{recall}_u = \frac{8}{8+5+3}$	$\textbf{recall}_n = \frac{60}{10+60+30}$	$\textbf{recall}_s = \frac{200}{1+50+200}$	

Figure 4.5 Confusion matrix for a three-class categorization task, showing for each pair of classes (c_1, c_2), how many documents from c_1 were (in)correctly assigned to c_2.

Evaluating with more than two classes

Now we have 3 separate precisions and 3 separate recalls… how do we combine them into 1?

Macroaveraging:

- compute the performance for each class, and then average over classes

Microaveraging:

- collect decisions for all classes into one confusion matrix
- compute precision and recall from that table.

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Evaluating with more than two classes

Now we have 3 separate precisions and 3 separate recalls… how do we combine them into 1?

Class 1: Urgent

	true urgent	true not
system urgent	8	11
system not	8	340

$$\text{precision} = \frac{8}{8+11} = .42$$

Class 2: Normal

	true normal	true not
system normal	60	55
system not	40	212

$$\text{precision} = \frac{60}{60+55} = .52$$

Class 3: Spam

	true spam	true not
system spam	200	33
system not	51	83

$$\text{precision} = \frac{200}{200+33} = .86$$

Pooled

	true yes	true no
system yes	268	99
system no	99	635

$$\text{microaverage precision} = \frac{268}{268+99} = \mathbf{.73}$$

$$\text{macroaverage precision} = \frac{.42+.52+.86}{3} = \mathbf{.60}$$

Figure 4.6 Separate confusion matrices for the 3 classes from the previous figure, showing the pooled confusion matrix and the microaveraged and macroaveraged precision.

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